

HOWARD PARK

Unity Developer | Network Engineer | AWS & Unity
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Hello! In 7 years, I've been a tech artist, network engineer, UX/UI developer and game programmer all through Unity.

I have a strong foundation in VR, AR, Mobile and PC games and start-up, corporate and indie environments.

Making interactive experiences is my passion and I'm constantly wearing multiple hats to collaborate effectively with others.

WORK EXPERIENCE

Meta — Technical Artist

December 2023 - December 2024

- **Created a sandbox tool on Unity** for sound engineers to optimize noise profiles in high and low traffic settings.
- **Live-service experience developing new seasonal content and asset creation tools** for Horizon Worlds (100k+ users)
- Closely collaborated with other departments to scope technical requirements for engineers, designers and artists.
- Optimized scene performance by optimizing shaders, models and textures for both the Meta Quest and desktop clients.

FiveMics — Unity Network Engineer

August 2022 - November 2023

- Collaborated directly with the CEO to define and deliver the multiplayer vision for a hip-hop themed card game.
- **Translated designs and wireframes into functional prototypes** for UX artists, designers and artists.
- Architected the game's backend using AWS to produce scalable servers across the US.
- **Wrote REST APIs** on API Gateway and Lambda functions to manage multiplayer sessions and server-side game logic for card play.
- Created **access control policies** for developers and users for security and facilitate testing and mock user flows.
- Integrated CloudFormation for **build automation** and scalable infrastructure updates.
- Full-stack developed **user onboarding and registration flows** using Cognito and DynamoDb.
- Produced clean documentation guiding other engineers how to use the architecture and scale it.

Ultraviolet — Senior Unity Developer

April 2021 - June 2022

- **Led a small team** of Unity developers to build a metaverse prototype for the Octi social media app.
- Collaborated with iOS engineers to integrate Unity builds with XCode and administer two-way communication.
- Collaborated with ML engineers to **implement analytics APIs** to the game to monitor player activity and retention.
- **Programmed all gameplay and networking** for two multiplayer games using Unity Mirror and AWS Lightsail to spin-up private servers for testing.
- **Employed interpolation techniques to synchronize animations, physics and client data** across the server.
- Developed functional **UI screens and prototypes** for the UX team to test AR camera features, tap gestures, and min-games.
- Increased community engagement by 20%, reaching #1 in the social media category for two weeks.

Evava VR — Unity Developer (AR)

September 2020 - March 2021

- Produced AR experiences for fashion districts including an AR Santa for the COVID holidays and a stomping game using interactive floor projections.
- **Programmed all AR gestures** and mechanics including body tracking, face tracking and object manipulation.
- **Mentored colleagues** and interns on proper Unity habits and tools.

Konami Digital Entertainment — Quality Assurance Tester

Dec 2019 - March 2020

- **Reported and tracked bugs** to developers and conducted console compliance tests for *Yu-gi-oh! Legacy of the Duelist: Link Evolution* and *Skelattack!*

Tandon Online — Unity Developer

June 2018 - August 2018

- Collaborated with a professor to build an online educational game for students to teach organizational behavior principles and reward strategic teamwork under controlled chaos.
- **Networked all server-client interactions** including physics, crane mechanics and P2P matchmaking lobbies.
- Implemented **interpolation strategies to synchronize states and positions** of objects across clients to ensure smooth gameplay.
- Conducted weekly pitches on several design iterations and mechanics with faculty to reach project goals within reasonable scope.

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Education

New York University: Tandon School of Engineering (2015 - 2019)

Major: B.S. Integrated Digital Media

Minor: Game Engineering

Accolades: Dean's List 2016-2017

Relevant Courses: Game Programming, Data Structures & Algorithms, Object Oriented Programming, Artificial Intelligence, Ideation & Prototyping

Skills

Languages: C#, C++, Python, Javascript, Typescript, PostgreSQL, Java, OpenGL, HLSL

Frameworks / Tools: AWS, Supabase, Pandas, Numpy, Scikit-learn, Selenium, BeautifulSoup

Engines/Libraries: Unity, Unreal, Gamemaker, SFML C++, PyGame

Projects

LoL Draft Simulator — Supabase, Network Architecture, Data Management/Analysis, Full-stack Unity

November 2024 - Present

- Developed a collectable card game prototype on Unity where players can collect cards of real League of Legends competitors, build synergistic rosters, and compete in a series of strategic ban-pick tournaments to gain rewards.
- **Processed user tokens** to create a secure registration and login flows.
- **Automated real-time data manipulation** using typescript to improve data integrity and operational efficiency.
- Wrote Python scripts to **programmatically scrape and import millions of player and game metrics** using the CargoQuery API.
- **Trained a logarithmic regression model** using sklearn to calculate roster strength and apply categorical weighting for strategic game modes and gimmicks.

Technologies used: Supabase, Typescript, Javascript, PostgreSQL, Python, Unity/C#

AR Waifu — Augmented Reality, AI, Pathfinding

June 2023 - July 2023

- One month exploration project to learn AR, AI and pathfinding algorithms. The product is an augmented reality management-sim where players oversee AI-driven anime characters under a house by maintaining their relationships, health and household tasks.
- Created **task queues and behavior trees** for each character to automate a schedule of tasks and behaviors based on personality, agitation and relationships with other characters.
- Implemented a grid-based **A* pathfinding algorithm** to maneuver characters around the house.
- Added touch gestures to move and engage characters with interactable objects and characters.

Technologies used: Unity/C#, ARFoundations, iOS SDK

"HMDR VR" and "Grounded" — VR, Unity

2018 - 2019

- **Programmed puzzles, controls and gameplay mechanics** on Unity for two narrative VR projects for game development classes.
- Designed a **narrative system** that tracked player progress and triggered changes in environment, sounds and level design for immersive gameplay.

Technologies used: Unity/C#, ARKit

Interests and Fun Facts

- I'm a competitive gamer, reaching high ranks in Marvel Rivals, TFT and Overwatch.
- My project car is an Acura RSX that I've fully restored with some friends as a hobby.
- I am working part-time as a barista to learn about coffee.
- I am learning Brazilian Jiu Jitsu and really bad at it.